Sysmon Event Log Analysis Report

1. Introduction

Sysmon (System Monitor) is a Microsoft Sysinternals tool used for monitoring detailed system activities in a Windows environment. It records important endpoint activities such as process execution, network connections, file changes, registry modifications, and DNS queries.

In this project, Sysmon was installed on a Windows VirtualBox machine and its logs were analyzed using Windows Event Viewer. The purpose of this project is to understand endpoint monitoring and basic SOC investigation techniques.

2. Project Objective

The main objectives of this project are:

- To install and configure Sysmon in a Windows environment.
- To understand Sysmon-generated security logs.
- To analyze important Sysmon Event IDs.
- To monitor system activities.
- To learn how SOC analysts investigate endpoint events.

3. Lab Environment

The project was performed in a controlled virtual lab environment.

Environment Details:

- Virtualization Platform: Oracle VirtualBox
- Operating System: Windows
- Monitoring Tool: Sysmon
- Log Analysis Tool: Windows Event Viewer
- Command Tool: PowerShell

4. Sysmon Installation

Sysmon was downloaded from Microsoft Sysinternals and installed with administrator privileges.

After successful installation, Sysmon started collecting system activity logs.

The installation was verified by checking the Sysmon service and Event Viewer logs.

5. Log Collection Process

Sysmon logs were accessed through Windows Event Viewer.

The log location was:

Applications and Services Logs → Microsoft → Windows → Sysmon → Operational

The operational log contains all Sysmon-generated security events.

6. Testing Method

PowerShell was used to generate normal system activities for testing.

The following applications were executed:

- Notepad
- Command Prompt
- PowerShell

After execution, the generated activities were captured by Sysmon and analyzed in Event Viewer.

7. Event ID Analysis

Event ID 1 - Process Creation

This event records when a new process starts in the system.

The analysis included:

- **Process name**
- **Process ID**
- **Parent process**
- **Command line**
- **User account**
- **File location**
- **Hash information**

This event is useful for detecting suspicious programs or unauthorized execution.

Event ID 3 - Network Connection

This event records network connections created by processes.

The observed information includes:

- **Source address**
- **Destination address**
- **Port number**
- **Protocol**
- **Process name**

SOC analysts use this event to identify suspicious communication and possible malware connections.

Event ID 5 - Process Termination

This event records when a running process stops.

It helps analysts understand the complete lifecycle of a process.

The collected details include:

- **Process name**

- **Process ID**
- **User**
- **Time information**

Event ID 7 - Image Loaded

This event records when DLL files or modules are loaded by processes.

It helps detect suspicious DLL loading behavior.

Event ID 11 - File Created

This event records the creation of new files.

The analysis includes:

- File name
- File path
- Creating process
- User information

This can help identify suspicious file drops.

Event ID 13 - Registry Modification

This event records changes made to Windows Registry values.

Registry monitoring is important because attackers often use registry changes for persistence.

Event ID 22 - DNS Query

This event records DNS requests made by applications.

It helps identify:

- Domain lookups
- Suspicious domains
- Application network behavior

8. Findings

During the analysis, Sysmon successfully captured different Windows activities.

The investigation showed:

- Process executions were recorded.
- Network activities were logged.
- File activities were monitored.
- Registry changes were captured.
- DNS queries were available.

These logs provide detailed visibility into endpoint behavior.

9. SOC Analyst Importance

Sysmon is valuable for SOC operations because it provides detailed endpoint telemetry.

It helps analysts to:

- **Investigate security incidents.**
- **Perform threat hunting.**
- **Detect suspicious activities.**
- **Analyze malware behavior.**
- **Support incident response.**

10. Skills Learned

Through this project, I learned:

- **Sysmon installation and configuration.**
- **Windows Event Viewer analysis.**
- **Endpoint monitoring concepts.**
- **Event ID investigation.**
- **PowerShell testing.**
- **Basic SOC investigation workflow.**

11. Conclusion

This project provided practical experience in Windows endpoint monitoring using Sysmon.

By analyzing Sysmon Event Logs, system activities can be monitored and suspicious behavior can be investigated. This knowledge builds a strong foundation for SOC Analyst roles and security monitoring tasks.